Circular Motion Revision Sheet

Level 2 Sheet

Note: Revise Class Notes + Do Last 5 Years of PYQ + These Sheets of Level 1 to 3.

1. A car moving with a speed of 54 km/h takes a turn of radius 20 m. A simple pendulum is suspended from the ceiling of the car. Determine the angle made by the string of the pendulum with the vertical during the turning. (Take $g = 10\ \text{m/s}^2$)

(A) $\tan^{-1}(0.5)$ (B) $\tan^{-1}(0.75)$

(C) $\tan^{-1}(1.125)$ (D) $\tan^{-1}(0.25)$

2. A car of mass 'm' moves on a banked road having radius 'r' and banking angle θ. To avoid slipping from banked road, the maximum permissible speed of the car is v_0. The coefficient of friction μ between the wheels of the car and the banked road is

(A) $\mu = \dfrac{v_0^2 + rg\tan\theta}{rg + v_0^2\tan\theta}$ (B) $\mu = \dfrac{v_0^2 - rg\tan\theta}{rg - v_0^2\tan\theta}$

(C) $\mu = \dfrac{v_0^2 - rg\tan\theta}{rg + v_0^2\tan\theta}$ (D) $\mu = \dfrac{v_0^2 + rg\tan\theta}{rg - v_0^2\tan\theta}$

3. A car of 800 kg is taking turn on a banked road of radius 300 m and angle of banking 30°. If coefficient of static friction is 0.2 then the maximum speed with which car can negotiate the turn safely: ($g = 10\ \text{m/s}^2$, $\sqrt{3} = 1.73$)

(A) 264 m/s (B) 51.4 m/s

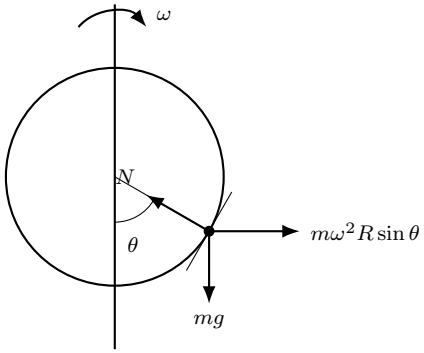
(C) 70.4 m/s (D) 102.8 m/s

4. A particle is moving in a circle of radius 50 cm in such a way that at any instant the normal and tangential components of its acceleration are equal. If its speed at $t = 0$ is 4 m s^{-1}, the time taken to complete the first revolution will be $\dfrac{1}{\alpha}\left[1 - e^{-2\pi}\right]$ s, where $\alpha =$ _____.

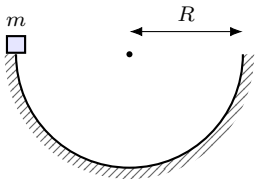
5. A motorcyclist wants to drive on the vertical surface of a wooden 'well' of radius 5 m, with a minimum speed of $5\sqrt{5}$ m/s. The minimum value of coefficient of friction between the tyres & the wall of the well must be

(1) 0.10 (2) 0.20 (3) 0.30 (4) 0.40

6. A thin circular loop of radius R rotates about its vertical diameter with an angular velocity ω. What is the angle made by the radius vector joining the centre to the small bead on the loop with vertical downward direction for $\omega = \sqrt{\dfrac{2g}{R}}$? (Neglect friction)



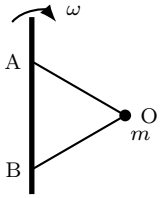
7. In the figure shown, a small block of mass $m = 2$ kg moves in fixed semicircular smooth track of radius $R = 2$ m in vertical plane. It is released from the top. Find the the resultant force on the block in newtons at the lowest point of track (Take $g = 10$ m/s^2)



8. A particle is moving with constant angular acceleration $= \alpha$ in a circular path of radius $\sqrt{3}$ m. At $t = 0$, it was at rest and at $t = 1$ sec, the magnitude of its acceleration becomes $\sqrt{6}$ m/s^2, then α is :

(1) 2 rad/s^2 (2) $\sqrt{3}$ rad/s^2 (3) $\sqrt{2}$ rad/s^2 (4) 1 rad/s^2

9. A particle O of mass m is attached to a vertical rod with two inextensible strings AO and BO of equal lengths ℓ. If ℓ be the distance between the points of suspension on the vertical rod, if the set up rotation with angular velocity ω, then



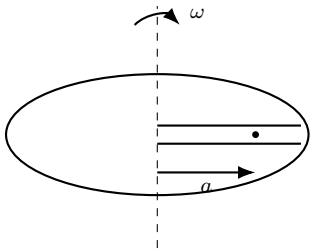
(1) Tension is thread AO is greater

(2) Tension is thread BO is greater

(3) Tension in the two threads are equal.

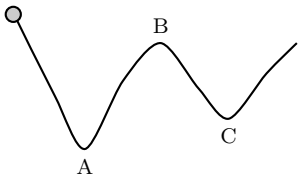
(4) Tension in AO or BO is greater according as ω is anticlockwise or clockwise.

10. A circular table with smooth horizontal surface is rotating with constant angular speed ω about its axis which is vertical and passing through its center. A groove is made on the surface along the radius and a particle is gently placed inside the groove at a distance 'a' from the centre of disc. Find the speed of the particle with respect to the table as its distance from the centre becomes ℓ. (Both a and ℓ are less then radius of circle and groove is smooth)



(1) $V = \omega\ell$ (2) $V = \omega(\ell - a)$ (3) $V = \dfrac{\omega(\ell + a)}{2}$ (4) $V = \omega\sqrt{\ell^2 - a^2}$

11. A body moves along an uneven surface with constant speed at all points. The normal reaction of the road on the body is :



(1) maximum at A (2) maximum at B (3) minimum at C (4) the same at A, B & C

Keep Going!

"Like an object in circular motion, stay centred on your goal and its pull will keep you on course."